

Computer Science I - CS 126

16 Weeks
Class



By Volodymyr (Vova) Saruta

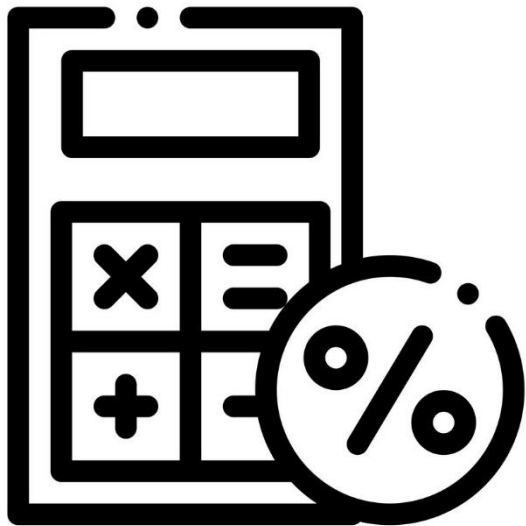
UNIT 1 PART 1

Computer Science?

Computer Science?

Computer Science

- Computer science is the study of **computation, automation, and information**



Computer Science

- CS is about PROCESS – describing how to accomplish tasks
 - "efficiently implementing automated abstractions"
([Philip Guo](#))
- Computers are a tool
 - Currently the best implementation platform
 - What kinds of problems can they solve?
 - How can they be made faster, cheaper, more efficient...?

Computer Science

- Science?
 - More like engineering, art, magic...
 - Hypothesis creation, testing, refinement important
- CS is still a young field finding itself



- program?
- program execution?
- programming language?

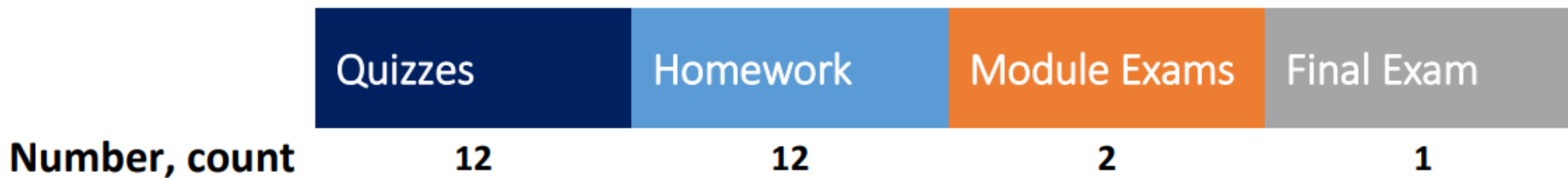
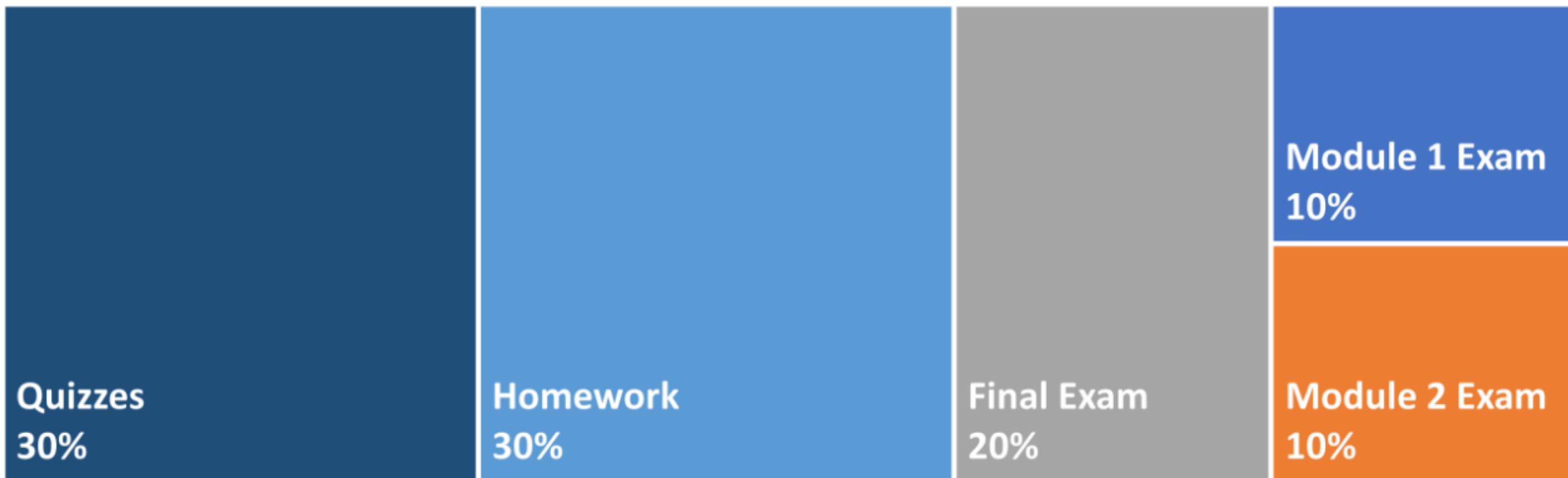


- **program:** A set of instructions to be carried out by a computer.
- **program execution:** The act of carrying out the instructions contained in a program.
- **programming language:** A systematic set of rules used to describe computations in a format that is editable by humans.



- *procedural languages*: programs are a series of commands
 - **Pascal** (1970): designed for education
 - **C** (1972): low-level operating systems and device drivers
- *functional programming*: functions map inputs to outputs
 - **Lisp** (1958) / **Scheme** (1975), **ML** (1973), **Haskell** (1990)
- *object-oriented languages*: programs use interacting "objects"
 - **Smalltalk** (1980): first major object-oriented language
 - **C++** (1985): "object-oriented" improvements to C
 - successful in industry; used to build major OSes such as Windows
 - **Python** (1991):
 - The language taught in this course

■ Module 1 Exam ■ Module 2 Exam ■ Final Exam ■ Quizzes ■ Homework



Monday	Lecture X.1 Coding Part X Assigned	
Tuesday		Coding Part (X-1).2 Due Quiz (X-1) Due
Wednesday	Lecture X.2	
Thursday	Quiz X Assigned	
Friday		Coding Part X.1 Due

Monday	Lecture 3.1 Coding Part 3 Assigned	
Tuesday		Coding Part 2.2 Due Quiz 2 Due
Wednesday	Lecture 3.2	
Thursday	Quiz 3 Assigned	
Friday		Coding Part 3.1 Due

Monday	Coding Part X Assigned	
Tuesday	Lecture X.1	Coding Part (X-1).2 Due Quiz (X-1) Due
Wednesday		
Thursday	Lecture X.2 Quiz X Assigned	
Friday		Coding Part X.1 Due

Monday	Coding Part 3 Assigned	
Tuesday	Lecture 3.1	Coding Part 2.2 Due Quiz 2 Due
Wednesday		
Thursday	Lecture 3.2 Quiz 3 Assigned	
Friday		Coding Part 3.1 Due

What would you learn/master?

- Problem solving and algorithmic thinking skills
- Software creation and maintenance
- Production of programmatic solutions
- Some Python

Join Discord for CS126



Why Python?

Why Python?

- Relatively simple
- Pre-written software
- Widely used

About Me

Academic experience

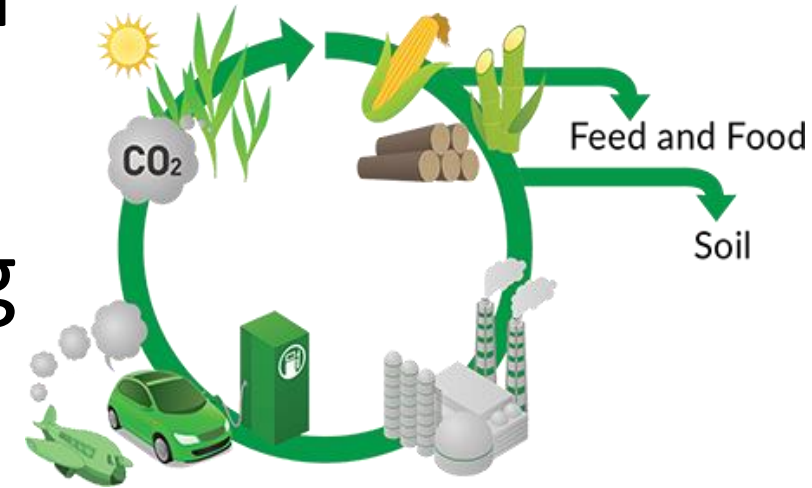
Achieved:

Masters in Ecology and Environmental Protection

Masters in Informatics and Computing

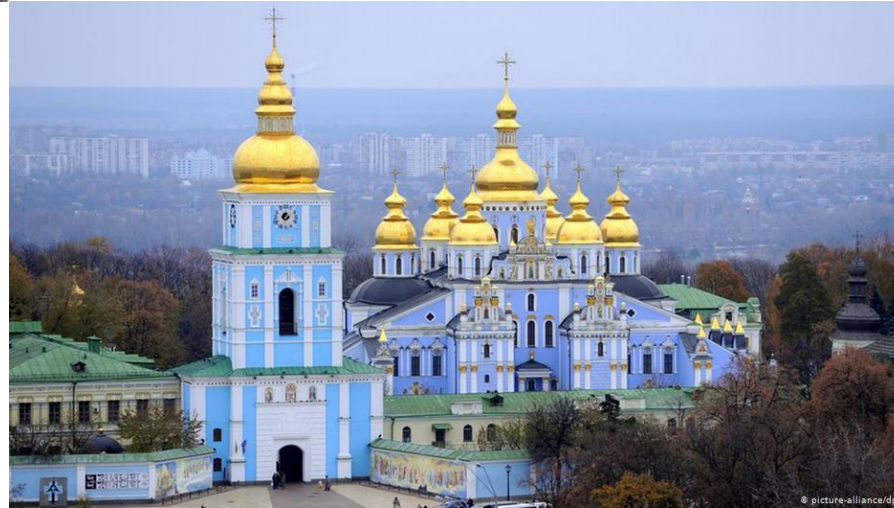
Currently:

Ph.D. Candidate in Informatics and Computing
with Ecological and Environmental emphasis



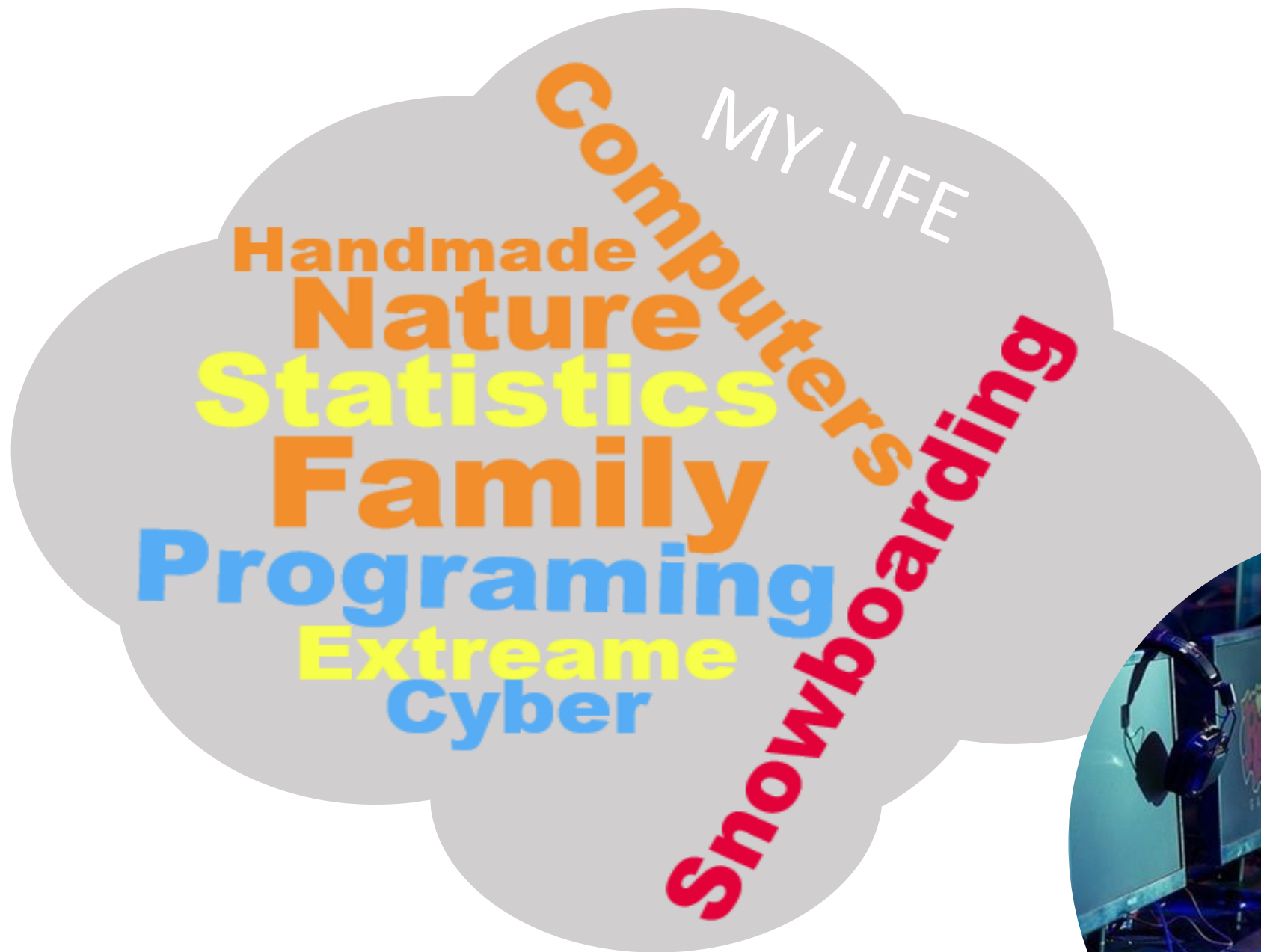


MY LIFE



UKRAINE | KYIV





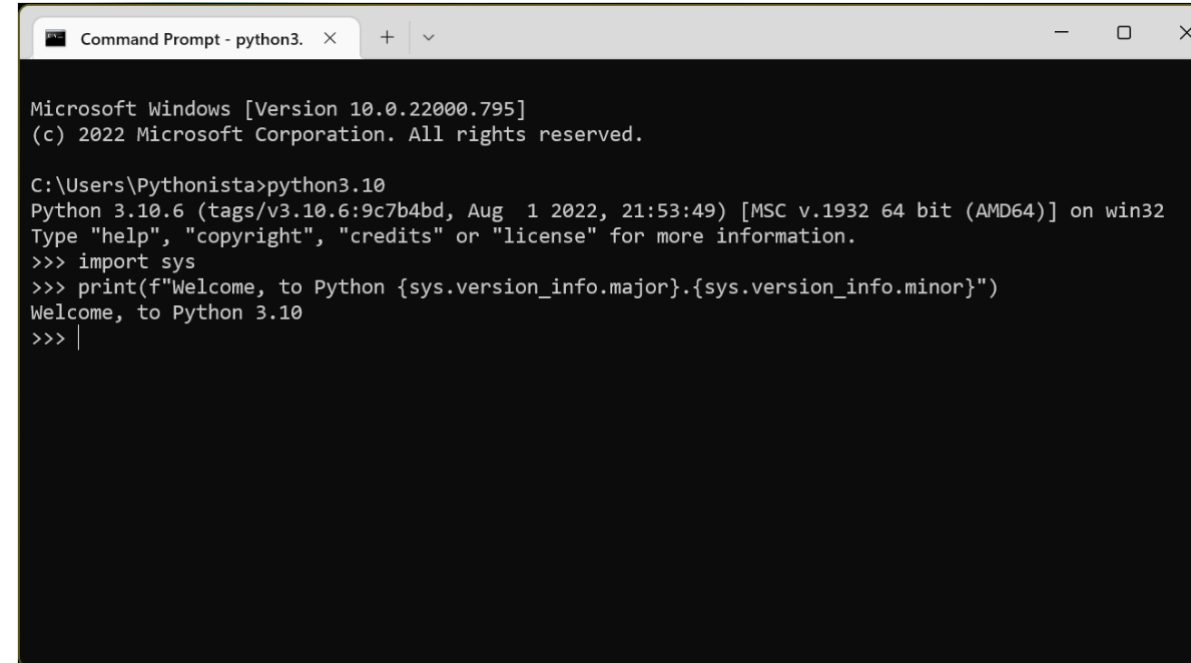
A Python program

```
print("Hello, world!")  
print()  
print("This program produces")  
print("four lines of output")
```

- Its output:

Hello, world!

This program produces
four lines of output

A screenshot of a Windows Command Prompt window titled "Command Prompt - python3.". The window shows the output of running a Python program. The text displayed is: "Microsoft Windows [Version 10.0.22000.795] (c) 2022 Microsoft Corporation. All rights reserved. C:\Users\Pythonista>python3.10 Python 3.10.6 (tags/v3.10.6:9c7b4bd, Aug 1 2022, 21:53:49) [MSC v.1932 64 bit (AMD64)] on win32 Type "help", "copyright", "credits" or "license" for more information. >>> import sys >>> print(f"Welcome, to Python {sys.version_info.major}.{sys.version_info.minor}") Welcome, to Python 3.10 >>> |". The prompt characters ">>>" indicate the Python interpreter's prompt.

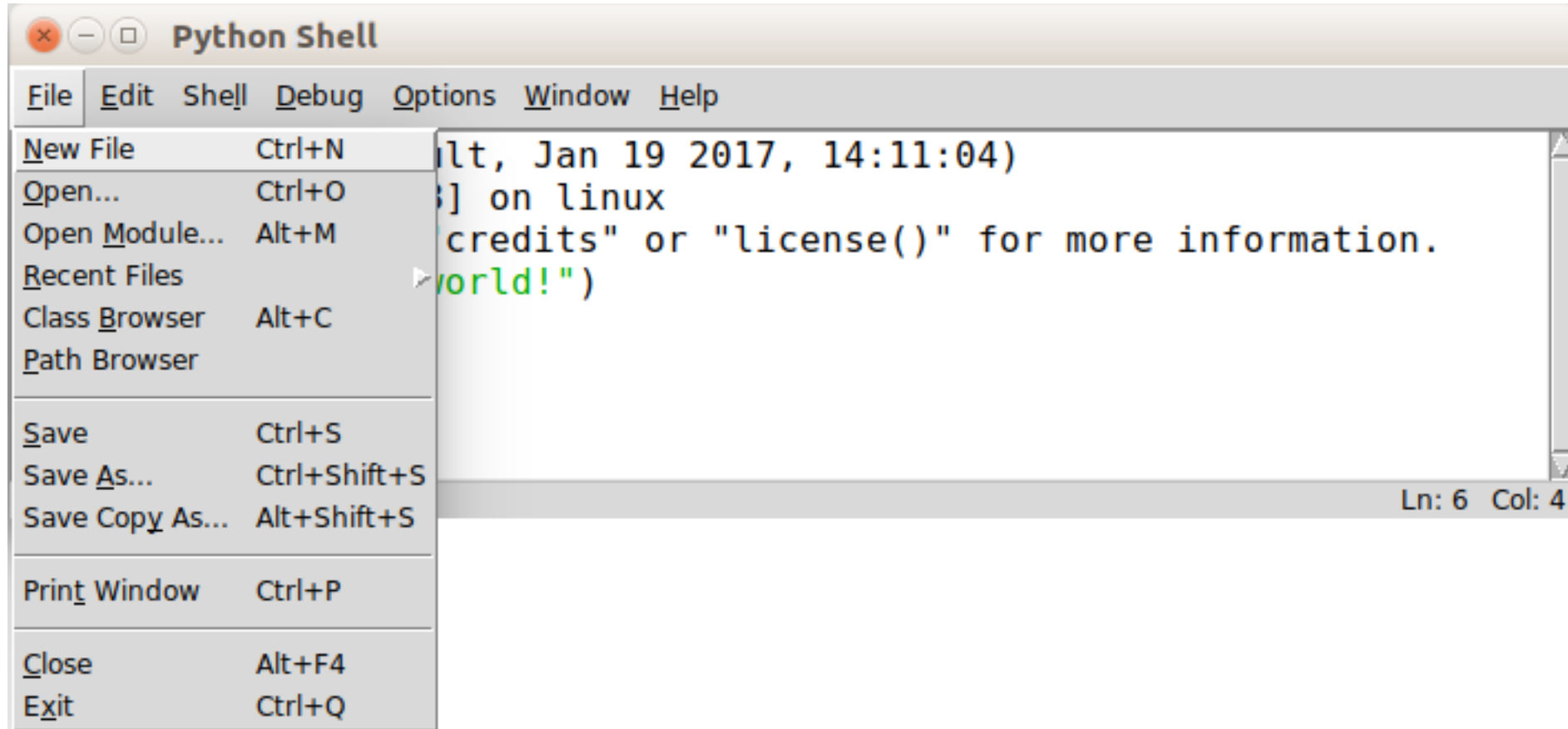
```
Command Prompt - python3. x + v  
Microsoft Windows [Version 10.0.22000.795]  
(c) 2022 Microsoft Corporation. All rights reserved.  
  
C:\Users\Pythonista>python3.10  
Python 3.10.6 (tags/v3.10.6:9c7b4bd, Aug 1 2022, 21:53:49) [MSC v.1932 64 bit (AMD64)] on win32  
Type "help", "copyright", "credits" or "license" for more information.  
>>> import sys  
>>> print(f"Welcome, to Python {sys.version_info.major}.{sys.version_info.minor}")  
Welcome, to Python 3.10  
>>> |
```


`print`

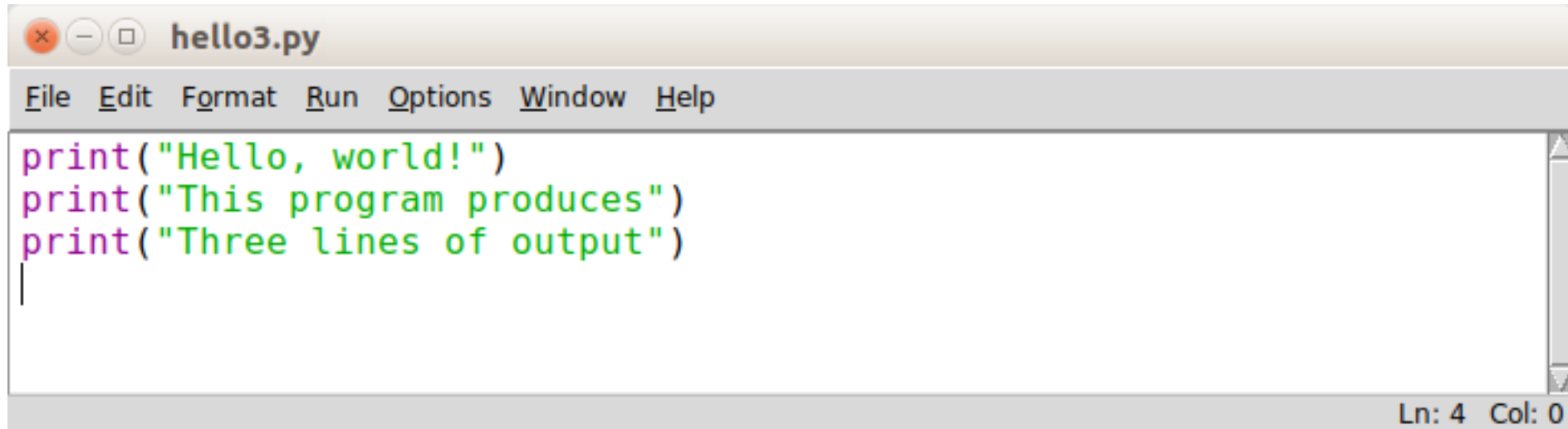
- A statement that prints a line of output on the console.
- Two commonly used ways to use `print` :
 - `print ("text")`
Prints the given message as output.
 - `print ()`
Prints a blank line of output.

Creating a Python Program

Creating a Python Program File



Creating a Python Program File

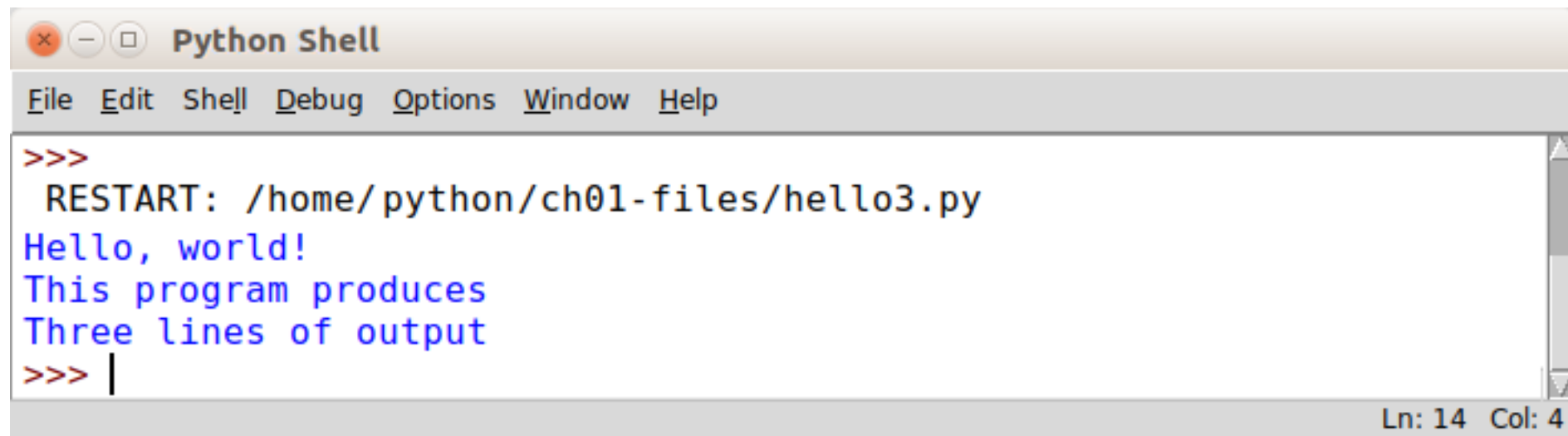


A screenshot of a Python IDE window titled "hello3.py". The window has a menu bar with "File", "Edit", "Format", "Run", "Options", "Window", and "Help". The main text area contains the following Python code:

```
print("Hello, world!")  
print("This program produces")  
print("Three lines of output")  
|
```

The status bar at the bottom right indicates "Ln: 4 Col: 0".

When Run -> Run Module is selected:



A screenshot of a Python Shell window titled "Python Shell". The window has a menu bar with "File", "Edit", "Shell", "Debug", "Options", "Window", and "Help". The main text area shows the execution of the script:

```
>>>  
RESTART: /home/python/ch01-files/hello3.py  
Hello, world!  
This program produces  
Three lines of output  
>>> |
```

The status bar at the bottom right indicates "Ln: 14 Col: 4".

Comments

- **comment:** A note written in source code by the programmer to describe or clarify the code.
 - Comments are not executed when your program runs.

- Syntax:

`# comment text`

- Examples:

`# This is a one-line comment.`

`# This is a very long
multi-line comment.`

Comments example

```
# Suzy Student,  
# CSc 110, Fall 2019  
# Displays lyrics  
  
# first line  
print("When I first got into magic")  
print("it was an underground phenomenon")  
print()  
  
# second line  
print("Now everybody's like")  
print("pick a card, any card")
```